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Whispers in the oil market: Exploring sentiment and uncertainty insights

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ABSTRACT

This paper develops a set of innovative text-based indices capturing oil market sentiment and oil price uncertainty. The textual analysis includes over 6 million news items spanning from January 1982 to June 2021. The evidence shows that sentiment indicators readily react to economic and geopolitical events that impact oil prices, thereby enabling these indicators to predict oil prices accurately. In contrast, uncertainty measures have inherent weaknesses, thus yielding unreliable oil price forecasts. This research yields a novel and robust text indicator that provides valuable insights for predicting the intricate dynamics of crude oil prices, particularly excelling in short-term forecasts and during periods of economic recession.

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1. Introduction

Despite growing investment in green alternatives, oil remains the world's most sought-after commodity due to its unparalleled versatility across various industries. Furthermore, fluctuations in crude oil prices ripple through the global economy, affecting key macroeconomic indicators, including inflation, consumer spending, and real GDP growth. Therefore, accurately predicting oil prices is not merely a matter of market speculation but a crucial necessity for central banks seeking to make informed policy decisions in an increasingly volatile economic landscape.

To this end, a variety of econometric models are proposed in the literature (see, e.g., Alquist & Kilian, 2010; Baumeister et al., 2015; Baumeister & Kilian, 2012, 2014, 2015). While these early models offer improved short- and medium-term forecasting capabilities compared to the no-change forecast, they show two significant limitations. First, most were calibrated under relatively stable market conditions, with oil price forecasting becoming notably challenging after 2010 (see Baumeister et al., 2022). Second, their long-horizon accuracy falls short of that of a simple random walk forecast. To address these challenges, Baumeister et al. (2022) propose a novel global economic conditions (GECON) indicator, enhancing short- and long-term oil price forecasts. However, despite these recent improvements, existing methods remain vulnerable to unforeseen geopolitical events impacting the oil market. One reason for this is that useful predictors are often made available with delays and require frequent revisions, preventing timely forecasts (Alquist et al., 2013; Baumeister & Kilian, 2012). Moreover, some of these variables are inherently slow to respond to key events that impact the price of oil.

Building on the limitations of existing models, this study introduces a novel text-based driver for the oil

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market that captures real-time sentiment insights previously overlooked. Constructed from oil-related news articles, the indicator is immune to data revisions. It reacts swiftly to events, such as geopolitical conflicts or supply-side decisions, which often elude slower-moving predictors. These properties ensure its stability and reliability, leading to substantial improvements in monthly real oil-price forecasts, which enhance point accuracy, density estimates, and directional precision, particularly during economic recessions.

The textual analysis[2] draws on over six million daily articles from *The Financial Times*, *Thomson Reuters*, and *The Independent* spanning from January 1982 to June 2021. News items are downloaded from the LexisNexis database, and are selected based on the joint occurrence of the words "oil" and "price". The semantic analysis is then conducted as follows.

Text information is initially gathered from *The Financial Times*, the longest available digital sample on the LexisNexis database. Fourteen text indicators are developed and integrated into various econometric models to forecast oil prices. The analysis is then extended by including articles from *Thomson Reuters* and *The Independent*, which are available for a shorter time horizon. The study explores which combination of these sources generates the most effective text indicators for forecasting various oil price measures. Following this, I extract the common component from the top-performing text indicators, resulting in substantial improvements in oil price forecasts across various time horizons. For point forecasts, the analyses reveal promising results, demonstrating statistically significant gains of 1% in the short run, whereas the medium- and long-term analyses indicate marginal gains of up to 10% compared to both models that exclude the text indicator and the random walk model. Similarly, density forecasts demonstrate strong statistical significance for short-, medium-, and long-term forecasts, outperforming models that do not include the text indicator.

This study makes two significant contributions. First, it introduces a novel text oil sentiment indicator (TOSI), constructed from a comprehensive, carefully curated collection of oil-focused news articles. Used alongside traditional oil market fundamentals, the proposed index significantly improves monthly forecasts of real oil prices. Robustness exercises further demonstrate that TOSI is not simply a proxy for general market sentiment or uncertainty. It correlates only modestly with conventional indices, yet reacts strongly to events directly affecting the oil market, such as geopolitical conflicts and oil production decisions. Second, the paper presents a comparative analysis of supervised text analytics techniques and various energy data sources, offering insights into the most effective text data for oil price forecasting. The results demonstrate that sentiment indicators respond strongly to oil market events, such as wars, natural disasters, and recessions, enabling accurate real oil price forecasts. In contrast, uncertainty series exhibit erratic swings that weaken their predictive power.

Related literature. Studies forecasting real oil prices present a broad and evolving literature. Among quantitative approaches, vector autoregression (VAR) models and their Bayesian extensions (BVARs) have been widely recognised for their superior short-horizon forecasting performance. In particular, Baumeister and Kilian (2012) demonstrate that recursive BVAR and standard VAR specifications consistently outperform Hotelling framework models, survey-based forecasts (Alquist & Kilian, 2010), standard autoregressive approaches and no-change forecasts up to six months ahead.

Subsequent works confirm and extend these findings. Alquist et al. (2013) demonstrate that both reduced-form and structural VAR models exhibit outstanding predictive ability, with structural specifications also enabling hypothetical forecasting scenarios beyond unconditional means. Baumeister and Kilian (2014) examine time-varying parameter VAR (TVP-VAR) models and conclude that, although they can address non-linearity in oil price movements, constant-parameter VARs yield the most accurate out-of-sample forecasts.

Another strand investigates combinations of forecasts. Baumeister et al. (2014) find that varying forecast combinations by horizon improves accuracy, while Baumeister and Kilian (2015) report that equally weighted averages of VAR-based forecasts, futures prices, no-change benchmarks and industrial raw materials outperform individual methods for horizons from one to eighteen months, with results robust up to six quarters ahead. Mixed-frequency approaches have also been tested: Baumeister et al. (2015) and Degiannakis and Filis (2018) employ MIDAS and mixed-frequency VAR models, respectively, with the latter finding gains from incorporating intra-day data.

An alternative literature champions structural macroeconomic models. Mănescu and Van Robays (2014) and Rubaszek (2021) demonstrate that DSGE models can match or exceed the point-forecast and directional accuracy of standard VAR and BVAR methods. Yet these comparisons typically omit stochastic-volatility BVARs (SVBVARs), which account for time-varying error variances and are now commonplace in applied macroeconomics (Clark, 2011). In this context, Baumeister et al. (2022) find that recursive forecasts through SVBVAR models deliver superior point and density forecasts, especially around major geopolitical events; however, their performance weakens during periods of acute economic instability due to the slow-moving nature of the underlying fundamentals. Table 1 summarises the key findings from these models.

To address these gaps, I propose a new text-based sentiment indicator within the SVBVAR framework, which substantially improves the accuracy of real oil price forecasting, particularly during periods of economic change. This study diverges from previous studies that employ text data for oil price analysis in both focus and methodology. While Wei et al. (2017) demonstrate the superiority of the Global Economic Policy Uncertainty index over country-specific EPU indices in forecasting crude oil

[2] In this paper I use the terms "text mining", "natural language processing (NLP)", and "text/textual analytics/analysis" interchangeably.

Table 1
Summary of key studies on real oil price forecasting.

Authors	Forecasting models	Predictors	Forecast frequency	Forecast horizon	Evaluation period	Most accurate model(s)
Alquist and Kilian (2010)	RW, FF, FSF, SF, HM	OF, OFS, SD	Monthly	1–12 months	1991M01–2007M02	RW
Baumeister and Kilian (2012)	RW, FF, AR, ARMA, VAR, BVAR	OF, IRM, BDI, OMF	Monthly	1–12 months	1992M01–2010M06	BAR, BVAR
Alquist et al. (2013)	RW, FF, AR, ARMA, VAR, BVAR	OF, OMF	Monthly	1–12 months	1991M12–2009M08	AR, VAR (short-term), RW (long-term)
Baumeister and Kilian (2014)	RW, FF, VAR, BVAR, TVP, PF	OF, OMF, ER	Monthly, Quarterly	1–12 months	1992M01–2011M06	monthly-VAR
Baumeister et al. (2014)	RW, VAR, FF TVP, PF	OF, IRM, OMF	Monthly, Quarterly	1–24 months	1992M01–2012M09	monthly-PF
Mănescu and Van Robays (2014)	RW, FF, DSGE, VAR, BVAR, PF	OF, OMF, CRB	Monthly, Quarterly	1–11 quarters	1995M01–2012M12	monthly-PF
Baumeister and Kilian (2015)	RW, FF, VAR, TVP, PF	OF, OMF, IRM	Monthly, Quarterly	1–24 months	1992M01–2012M09	monthly-PF
Baumeister et al. (2015)	RW, VAR MF-VAR, MIDAS	OF, GS, BDI, FFR, LIBOR, CRB, ER	Monthly	1–24 months	1992M01–2012M09	MIDAS
Degiannakis and Filis (2018)	RW, AR, ARMA, VAR, BVAR, MIDAS	OMF, TBT	Monthly	1–12 months	2011M12–2015M08	MIDAS (short-term), BVAR (long-term)
Rubaszek (2021)	RW, AR, BAR, VAR, BVAR, DSGE	OMF, FFR	Quarterly	1–20 quarters	2008M01–2018M03	AR, DSGE
Baumeister et al. (2022)	VAR, BVAR, SVBVAR, PF	OMF, GECON, GSP	Monthly	1–24 months	1992M01–2018M08	SVBVAR

Note: In column 2 and 7, RW: random walk, FF: futures-based forecast, FSF: futures spread-based forecast, SF: survey-based forecast, HM: Hotelling model, AR: autoregressive models, BAR: Bayesian autoregressive models, ARMA: autoregressive moving average models, VAR: vector autoregressive model, BVAR: Bayesian VAR, SVBVAR: stochastic volatility BVAR, TVP: time-varying parameter models, PF: pooled forecast. In column 3, OF: oil futures, OFS: oil futures spreads, SD: survey data, IRM: industrial raw material, BDI: Baltic Dry Cargo Index, OMF: oil market fundamentals (global crude oil production, oil inventories, proxy for global real activity), ER: exchange rates, CRB: Commodity Research Bureau index, FFR: federal funds rate, GS: gasoline spread, TBT: tick-by-tick (financial and commodity) data, GECON: global economic condition indicator, GSP: global steel production.

volatility, this paper shows that sentiment-based indices deliver significantly better predictive performance, emphasising the limitations of relying solely on uncertainty measures. Moreover, the analysis extends the text-based forecasting literature (Li et al., 2019; Bai et al., 2022; Gong et al., 2022) by using the full text of leading financial newspapers rather than relying on headlines or brief online snippets. I examine monthly oil price changes from January 1982 to June 2021. This extended temporal horizon and richer textual dataset capture structural shifts in global demand, long-term supply adjustments and evolving policy debates. These elements facilitate a more comprehensive understanding of the macroeconomic context underlying oil price movements, thereby distinguishing this article's contribution from prior research.

The remainder of this paper is organised as follows. Section 2 describes the selection of oil-related articles. Section 3 details the methodologies used to construct sentiment and uncertainty indicators. Section 4 presents empirical applications of these text series to forecast alternative oil-price measures, including comparisons across different news-data sources. Section 5 introduces a novel text-based oil sentiment index, generated by combining the text series with the strongest predictive performance, and evaluates its effectiveness in forecasting real oil prices. Finally, Section 6 concludes.

2. Text data

Over 6 million daily news items, featuring articles from *The Financial Times*, *Thomson Reuters*, and *The Independent*, are retrieved from the LexisNexis database. Newspaper articles are then selected based on the joint occurrence of the words "oil" and "price" and are filtered for subjects related to the crude oil sector. A scanning-word analysis is conducted on the entire dataset as the words "oil" and "price" can also occur in articles unrelated to crude oil. For example, articles reporting news about olive oil, albeit correlated with petrol prices, do not directly affect the sentiment of oil investors. Such unrelated articles are excluded. The search command used to query the dataset is illustrated in Appendix B.

The selected articles include economic, political, and business news covering the global oil industry, as well as statements and opinions from key figures in oil companies. Duplicate articles with identical text content, issued on the same day but with different editions (e.g. USA, Asia) are excluded. I also disregard tables containing numerical data as they add little to the meaning of the textual content.

The text sources used in this study are chosen for their unbiased and highly factual reporting, as rated by the Media Bias/Fact Check agency. Favouring one political party over others when reporting news can indeed deviate from

Table 2
Data source descriptive statistics.

				average monthly audience				
	n. articles	articles/month	time period	UK	EU	Americas	RoW	Readership
Financial Times	2,857,111	6,028	1982M01-2021M06	32%	22%	32%	14%	22M
Independent	2,957,986	6,241	1988M09-2021M06	37%	10%	41%	12%	68M
Thomson Reuters	207,914	439	2002M11-2021M06	28%	35%	33%	4%	35M
Total	6,023,011	12,708	–	–	–	–	–	125M

the objective view of reality, and consequently, the integrity of a text-based indicator diminishes. Moreover, daily news from such data sources is read by users across multiple territories worldwide, ensuring that the information extracted from articles represents a robust, general perspective. The descriptive statistics of the newspapers and the associated monthly global reach are summarised in Table 2. The gross audience breakdowns shown in the right panel of Table 2 refer to 2021.

3. Text analytics

Standard methodologies discussed in Gentzkow et al. (2019) are used to preprocess articles and make text data suitable for algorithmic interpretation.[3] A set of algorithm-based text metrics is then used to generate ten indices designed to capture oil market sentiment and four reflecting oil price uncertainty. The former aim to track the interaction between oil prices and global economic growth, while the latter are expected to monitor the uncertainty in the oil market.

The rationale for employing NLP methods trained on pre-existing rules is twofold. Firstly, such metrics do not violate the spirit of a real-time out-of-sample forecast, as text indicators are re-computed (and re-normalised) at each forecasting step. Therefore, information about the future is excluded, thereby preventing data leakage. Secondly, algorithm-based metrics enable the construction of novel and unique time series functioning as indicators for oil price forecasts. Unsupervised learning methods are not employed as they would make different predictions with each training, even when trained on the same dataset.

In all the cases considered, k is a numerical vector of dimension $N \times 1$ representing an individual article. For a given month m, N is the number of unique words ω used, and K is the numerical array representing the monthly text data collection.

The simplest text mining approach investigated is the *unigram* count probability model, where the word-specific ω_i maximum likelihood estimation

$$p(\omega_i|k) = \frac{counts(\omega_i|k)}{\sum_{n=1}^{N} counts(\omega_n|k)}, \qquad \text{for } k = 1, \ldots, K, \tag{1}$$

is used for computing either a sentiment or an uncertainty indicator. Following Kalamara et al. (2022), words are stemmed and a sentiment proxy is estimated by counting the strings "`econom`". In contrast, the uncertainty proxy focuses on the count of the strings "`uncertain`". A more advanced statistical methodology used to generate an uncertainty index is the Boolean Extract command, where a count is made if the strings "`econom`" and "`uncertain`" jointly occur in the same article.

Dictionary methods are also explored. These metrics are used to develop sentiment indicators by assigning positive, negative or neutral values to words based on predefined lists of vocabulary with preallocated scores. The dictionaries that I employ are (1) Financial Stability by Correa et al. (2017), (2) Financial Liability by Loughran and McDonald (2011), (3) Afinn by Nielsen (2011), and (4) Harvard-IV.

Furthermore, oil market sentiment and oil price uncertainty are also estimated through the application of two geometrical models: (i) term frequency matrix (tfm) and (ii) term frequency - inverse document frequency matrix (tf-idf). Both strategies rely on the frequency of occurrences of stems used in the unigram count probability model. However, in contrast to Eq. (1), geometrical models are not proportionally sensitive to term frequency changes. Specifically, tfm is given by

$$tfm_{\omega_{i|k}} = \frac{\sum_{k=1}^{K} \omega_{i|k}}{\text{Total No. Doc.}}, \qquad \text{for } k = 1, \ldots, K, \tag{2}$$

and incorporates information about the frequency of a token in each article for the overall monthly data collection K_m. While tf-idf is given by

$$tf\text{-}idf_{\omega_{i|k}} = \frac{log\left[1 + tfm_{\omega_{i|k}}\right]}{log\left[1 + (K/n)\right]}, \qquad \text{for } k = 1, \ldots, K, \tag{3}$$

and helps to overtake the limits associated with the weighting scheme of Eq. (2). K is the number of articles in month m, and n is the number of articles in which the stem occurs.

The last three metrics used to detect oil market sentiment are VADER (Hutto & Gilbert, 2014), Opinion Sentiment and BERT (Devlin et al., 2018). Although all are rooted in computer science literature, they build on distinct principles. VADER is a rule-based approach that accounts for the intensity of sentiment and contextual nuances, such as punctuation and capitalisation. Opinion Sentiment is a machine learning approach that uses a support vector machine classifier to categorise words into positive and negative classes. BERT is a hybrid methodology that combines rule-based (pre-trained weights) and machine-learning (transformer architecture) approaches for achieving contextual understanding. More details on textual metrics are provided in Appendix C.

The monthly score is determined by the average of all article scores for that month. Such text indicators, along with oil-market fundamentals, are then used to forecast monthly oil prices. As no single indicator outperforms

[3] More in-depth information on text cleaning is provided in Appendix B

across every horizon and price measure, I extract the common component from the indices with the lowest in-sample average forecast errors. This co-movement series then serves as a real-time indicator in forecasting exercises.

4. VAR model forecast

This section examines whether incorporating text indices of oil market sentiment and oil price uncertainty into alternative SVBVAR models enhances forecasts of real oil prices. Forecast performance is measured by the ratio of each model's mean-squared prediction error (MSPE) to those of two benchmarks, a driftless random-walk[4] and the equivalent VAR without text, so that values below one indicate superior accuracy relative to each benchmark. Statistical significance of MSPE differences is assessed via the Diebold–Mariano test, which is recommended for both point and density forecast evaluations (see, e.g., Chan, 2017; Chan et al., 2024), and is valid for nested models (Carriero et al., 2015). Forecasts at horizons of 1, 3, 6, 12 and 24 months are produced using a recursively expanding-window scheme from January 1982 through June 2021. Estimation begins with the 1982 M1-2000 M12 sample. Parameters are re-estimated each month, and since real oil prices enter the VAR in log levels, predicted values are exponentiated for evaluation.

4.1. Bayesian inference with stochastic volatility

Crude oil prices exhibit pronounced periods of time-varying volatility, driven by geopolitical shocks, economic crises and policy decisions, making accurate forecasts difficult with constant-variance models. To capture these dynamics, I explore the integration of text variables introduced in Section 3, within the SVBVAR framework proposed by Carriero et al. (2019). The baseline specification is

$$Y_t = c + \Phi_1 Y_{t-1} + \cdots + \Phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \Sigma_t) \qquad (4)$$

where Y_t denotes the vector of 4×1 dependent variables, and Σ_t is a full covariance matrix with time-varying elements. The observed variables included in Y_t are the percentage change in global crude oil production, a proxy for the global real economy, the log of the real price of crude oil, and an estimate of the change in global crude oil inventories. For real oil prices, I consider the West Texas Intermediate (WTI) and Brent crude oil prices, each deflated by the U.S. consumer price index. Baumeister and Hamilton (2019)'s world industrial production index (WIP) is used as a measure of global economic activity. Without loss of generality, it is possible to assess

$$\Sigma_t = \left(A^{-1}\right) \Lambda_t \left(A^{-1}\right)' \qquad \text{and} \qquad \Sigma_t^{1/2} = \left(A^{-1}\right) \Lambda_t^{1/2}$$

with Λ_t being a diagonal matrix and $\lambda_{i,t}$ a generic diagonal element, for $i = 1, \ldots, N$ representing the ith dependent variable. A^{-1} is a lower diagonal matrix with ones on the main diagonal. The system in Eq. (4) can be then rewritten as

$$Y_t = c + \Phi_1 Y_{t-1} + \cdots + \Phi_p Y_{t-p} + A^{-1} \Lambda_t^{1/2} \epsilon_t, \quad \epsilon_t \sim iid\, N(0, I_N) \quad (5)$$

$$\text{with } \Lambda_t = \begin{pmatrix} \exp(\lambda_{1,t}) & 0 & \cdots & 0 \\ 0 & \exp(\lambda_{2,t}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \exp(\lambda_{N,t}) \end{pmatrix}, \qquad (6)$$

where $\lambda_{1,t}, \lambda_{2,t}, \ldots, \lambda_{N,t}$ refer to the log volatility of the N structural shocks in Eq. (4), and its law of motion is

$$\tilde{\lambda}_t = \tilde{\lambda}_{t-1} + \nu_t,$$

where $\tilde{\lambda}_t = \left[\lambda_{1,t}, \lambda_{2,t}, \ldots, \lambda_{N,t}\right]$ and $\nu_t \sim iid\, N(0, Q)$ with full variance matrix Q. Considering that the covariance matrix of the error term ϵ_t is diagonal, the estimation of the SVBVAR system in (5) can be done one equation at a time, leading to a vast reduction in the computational burden.

As a first exercise, I compare benchmark forecasts from Eq. (5) with those that include a text-based variable as the fifth endogenous predictor in the SVBVAR model. As outlined in Table 3, text regressors can improve the out-of-sample forecast of monthly oil prices significantly. In particular, there is evidence that computer science text metrics outperform the other metrics at any time horizon h. Specifically, BERT's forecasts are 1% statistically significant for WTI and 5% for Brent prices in the short run. In the medium term, BERT generates up to a 10% marginal improvement over the no-change forecast, while VADER is the best performing metric for the long run.[5]

An equivalent experiment is also performed using the SVBVAR model without text data as the benchmark, replacing the random walk. Each TOSI-based model's MSPE is divided by the MSPE of the no-text model under identical sample restrictions, yielding results that are fully consistent with the random-walk baseline. Sentiment-based measures outperform the no-text SVBVAR in the short term, while the computer science text metrics also deliver superior performance in both the medium and long term. Notably, while the random walk comparison yields statistically significant improvements only for one-step-ahead forecasts, the no-text benchmark reveals statistically significant gains of 1% in the short run and up to 10% in the medium and long run.

In a second experiment, I compare the difference between the cumulative sum of forecasting error (CSFE) of a text and a no-text SVBVAR. The results shown in Fig. 1 indicate that the strong performance of certain text metrics is consistent across short, medium and long terms. Moreover, there is evidence suggesting that text models perform well during periods of high volatility. Similar conclusions cannot be drawn when text regressors

[4] This strategy has long been used in the forecasting literature for real oil prices, as the random walk benchmark is notoriously difficult to beat (see, e.g., Alquist et al., 2013; Garratt et al., 2019; Hamilton, 2009, and more recently Baumeister et al., 2024).

[5] I also explore the alternative specification with text factors included as exogenous variables; those results are reported in Appendix H. However, the endogenous specification yields the highest forecasting accuracy.

Table 3
Recursive MSPE ratios for real oil price forecasts: SVBVAR(12) models with alternative text-based indicators relative to a random walk.

	h = 1		h = 3		h = 6		h = 12		h = 24	
Text index	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
No - text	0.881**	0.888**	0.909	0.917	0.935	0.937	0.994	0.982	0.930	0.919
Sentiment count	0.883**	0.897**	0.925	0.942	0.943	0.975	1.011	1.013	0.938	0.932
Uncertainty count	0.896**	0.911*	0.926	0.943	0.963	0.969	1.019	1.041	0.936	**0.917**
Boolean count	0.888**	0.905*	0.917	0.932	0.948	0.943	0.995	0.997	0.939	0.923
Financial Stability	**0.879****	0.890**	0.935	0.961	0.993	1.017	1.041	1.053	0.991	0.978
Financial Liability	0.871**	**0.884****	0.912	0.940	0.935	0.962	**0.959**	**0.977**	0.950	0.953
Afinn	0.860**	**0.880****	**0.898**	0.920	**0.927**	**0.932**	**0.978**	0.988	**0.923**	0.939
Harvard-IV	**0.878****	0.891*	**0.907**	0.936	0.944	0.962	1.006	1.048	0.940	0.934
Sentiment tfm	**0.877****	0.876**	**0.896**	0.904	**0.923**	**0.920**	**0.974**	**0.977**	0.948	0.924
Uncertainty tfm	**0.880****	0.895*	0.911	0.927	0.942	0.937	**0.989**	1.022	0.943	0.935
Sentiment tf-idf	**0.875****	**0.886****	0.910	0.926	0.955	0.963	1.047	1.034	1.016	0.983
Uncertainty tf-idf	**0.878****	**0.885****	0.922	0.921	0.939	0.947	**0.982**	1.010	0.959	0.948
VADER	0.864**	**0.883****	**0.895**	**0.914**	0.939	0.938	1.019	1.034	**0.914**	**0.902**
Opinion Sentiment	0.870**	**0.882****	0.913	**0.909**	**0.929**	0.915	**0.962**	0.986	0.949	**0.913**
BERT	0.834***	0.857**	0.870	**0.890**	0.901	0.901	**0.941**	0.925	0.940	0.925

Note: In column 1, the No-text row corresponds to the baseline (Baumeister et al., 2022)'s SVBVAR model, which does not incorporate any text-based indicators. The remaining rows present results where different text regressors are endogenised within the original SVBVAR framework. The bold values report that MSPEs of text-based SVBVARs outperforming MSPEs of no-text-based SVBVAR for each specific time horizon h and oil price measure (WTI and Brent). *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively, when tested against the random walk model, while blue, red, and yellow cells denote significance levels of 10%, 5%, and 1%, respectively, when tested against the No-text-based SVBVAR model. Significance tests are performed using the Diebold–Mariano test.

are employed to account for uncertainty in the oil market (see Fig. 2).

This evidence aligns with the responsiveness of time series text indicators to world oil market events. Indices reflecting oil market sentiment respond to more historical events impacting the crude oil industry compared to those capturing oil price uncertainty. Specifically, indices developed through dictionary- and computer science-based methodologies react to key historical events – such as the First Gulf War, the collapse of the Soviet Union, the Asian economic crisis, the Iran-Iraq War, the global financial crisis, the Venezuelan protests, and COVID-19 – and uniquely identify other events positively influencing oil market sentiment. For example, BERT captures both the clean-up operations following the Sea Empress oil spill in February 1996 and the Deepwater Horizon oil spill in April 2010, which significantly impacted the British economy and environment.

In contrast, there are instances where unigram count indicators fail to represent the impact of certain events on oil prices accurately. For example, the 1.5 million barrels per day cut that OPEC agreed in January 2001 is unlikely to cause a larger variation than the COVID-19 pandemic. Likewise, Boolean count does not perform as expected across the global financial crisis period and the 9/11 terrorist attack. Similar patterns are observed with uncertainty tfm and tf-idf, as both report an unexpected 13% increase in May 2000 despite a stable oil market. This lack of responsiveness is common in crude oil markets where uncertainty is typically driven by geopolitical events rather than global-specific financial disruptions. Such peculiarity is also identified by other oil price uncertainty indices developed through more structured data. See, for example, the conditional standard deviation indicator of Elder and Serletis (2010). Wars in oil-producing countries can lead to the destruction of oil fields, resulting in shortfalls in crude oil production and subsequent increases in oil prices (Edelstein & Kilian, 2009; Kilian, 2009). Therefore, the irrational reactions to geopolitical events may explain why uncertainty-based indicators, despite offering slight improvements in short-term forecasts, fail to perform well overall in predicting oil prices.

A more balanced representation of positive and negative events is provided by sentiment indicators developed through geometrical models. These indices respond to both positive events, such as the Iran-Iraq war ceasefire and the inauguration of oil shipments from Kazakhstan's Tengiz field, and negative events, such as the Venezuelan protests in June 2016, indicating their potential for predicting future oil price trends.

To further validate these findings, the analysis is also extended to include density forecasts, in accordance with the methodology outlined in Section 5.1. Density forecasts are compared to both a random walk benchmark and the equivalent model without text data. The evidence suggests that, compared to a random walk, both sentiment and uncertainty measures substantially improve forecast performance. Moreover, compared to the model without text, sentiment data consistently generate statistically significant improvements across short, medium, and long horizons; particularly for Brent prices (see Appendix H).

4.2. Are more sources of text data better?

Several studies conduct sentiment analysis relying on a single newspaper, implicitly assuming that one outlet captures the full spectrum of market sentiment (see, e.g., Bybee et al., 2024; Garcia, 2013; Tetlock, 2007). To relax this assumption and mitigate source bias, I add articles from *The Independent* and *Thomson Reuters* to the *Financial Times* corpus. For each outlet, I generate the same text indicators introduced earlier, then pool all oil-related

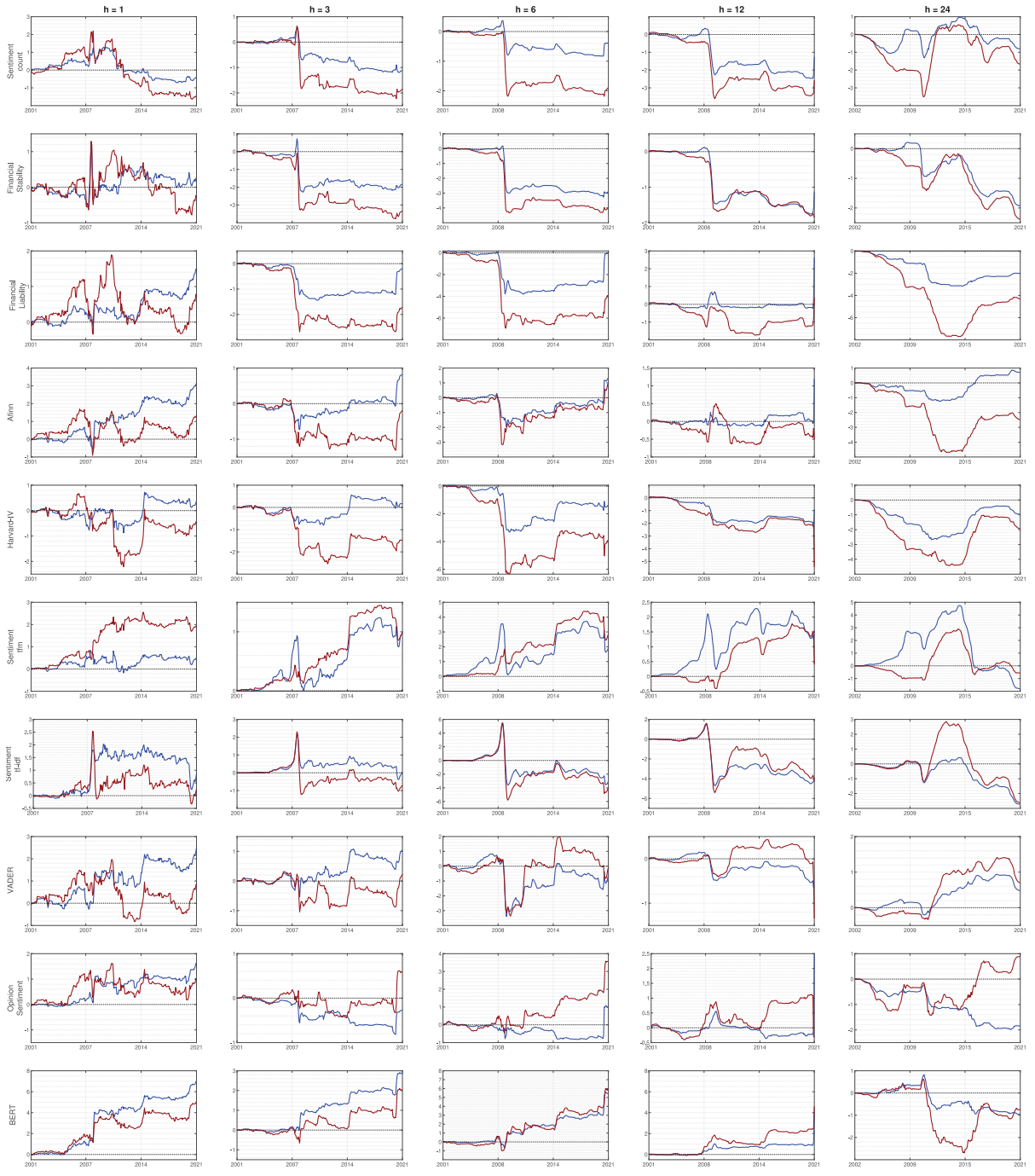


Fig. 1. The figure plots the difference in cumulative sum of forecasting errors between a 4-variable SVBVAR (without text indicators) and a 5-variable SVBVAR (with endogenised text indicators). Text variables account for oil market sentiment. The blue and red lines represent WTI and Brent crude oil prices, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

articles, rather than averaging by source, and compute a single sentiment index over three subperiods determined by archive availability. I evaluate every possible combination of these sources to isolate the incremental value of each. The resulting combined indices capture additional content that would otherwise be omitted, such as oil companies' CEO statements, shareholder updates, and plans of energy sector leading firms, including Laredo Petroleum, Nordic American Tankers, Seadrill, and Statoil, among many others. For exposition's sake, only results

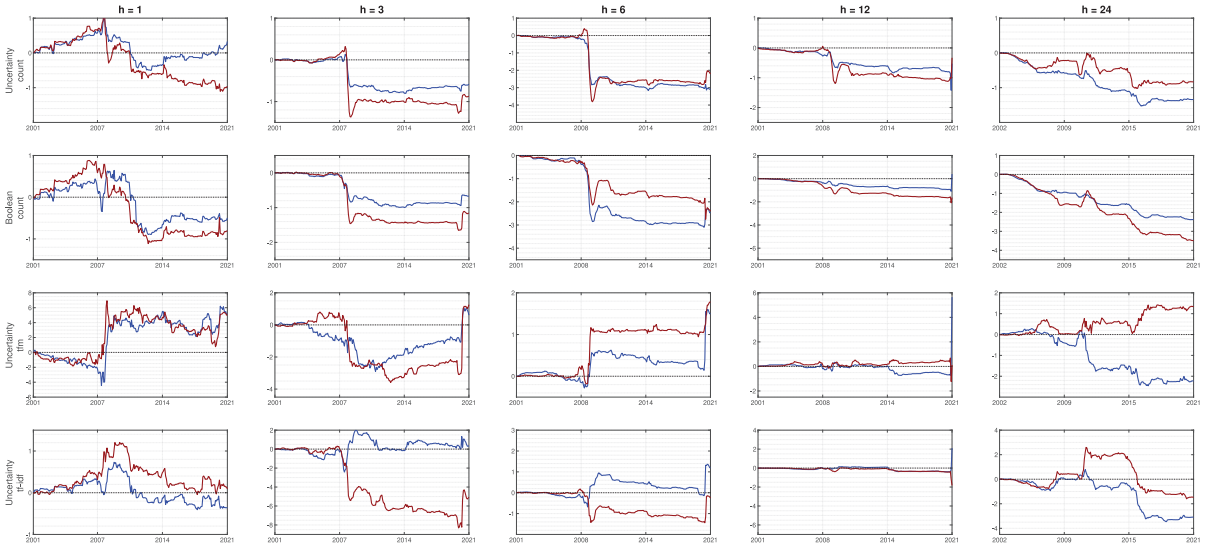


Fig. 2. The figure plots the difference in cumulative sum of forecasting errors between a 4-variable SVBVAR (without text indicators) and a 5-variable SVBVAR (with endogenised text indicators). Text variables account for oil price uncertainty. The blue and red lines represent WTI and Brent crude oil prices, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Recursive MSPE ratios relative to a random walk forecast of alternative monthly indicators of real oil prices. Combination between different text data sources.

	h = 1		h = 3		h = 6		h = 12		h = 24	
Text sources	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
	VADER									
FT-TR	0.858**	0.885**	0.886	0.916	0.915	0.940	0.983	1.002	0.898	0.884
FT-IND	0.865**	0.883**	0.901	0.925	0.939	0.938	1.004	1.007	0.912	0.897
FT-IND-TR	0.865**	0.883**	0.894	0.928	0.931	0.941	0.988	0.985	0.899	0.885
	Opinion Sentiment									
FT-TR	0.870**	0.881**	0.908	0.925	0.933	0.917	0.981	0.958	0.938	0.908
FT-IND	0.875**	0.891*	0.895	0.906	0.906	0.893	0.923	0.940	0.918	0.902
FT-IND-TR	0.869**	0.884**	0.893	0.903	0.905	0.884	0.946	0.941	0.951	0.939
	BERT									
FT-TR	0.829***	0.831***	0.858	0.861	0.897	0.881	0.935	0.922	0.926	0.911
FT-IND	0.848**	0.867**	0.878	0.895	0.902	0.898	0.944	0.921	0.934	0.915
FT-IND-TR	0.853**	0.865**	0.878	0.887	0.909	0.899	0.961	0.913	0.943	0.930

Note: In column 1, FT: Financial Times, TR: Thomson Reuters, IND: Independent. Blue values report the lowest MSPE results relative to a specific time horizon h and oil price measure (WTI and Brent). *, **, *** respectively denote 10%, 5% and 1% level of significance as suggested by the Diebold–Mariano test.

generated by text indices developed through computer science-based metrics are reported.

Table 4 reports the forecasting outcomes for the SVB-VAR models combining multiple news sources. Improvements are consistent across horizons and statistically significant at short horizons: the 1-month-ahead Brent forecast becomes significant at the 1% level, WTI's MSPE reduction increases slightly from 16.6% to 17.1%, and long-run MSPE reductions reach 10.2% for WTI and 11.6% for Brent. These findings underscore the value of integrating diverse, politically unbiased news feeds to enhance oil-price forecasting. These findings underscore the value of integrating diverse, politically unbiased news feeds to enhance oil-price forecasting.

5. A new text oil sentiment indicator

To distil the forecasting power of multiple sentiment measures into a single, timely index, I follow Alquist et al. (2013) and Hamilton (2021) by identifying those series that Granger–cause real oil prices and exhibit statistically significant p-values over the in-sample period January 1982–December 2000. The series meeting this criterion are VADER and BERT, both constructed from multiple news sources. I then extract the first principal component of these two standardised series to form the Text Oil Sentiment Indicator (TOSI), which captures 95% of their joint variation.

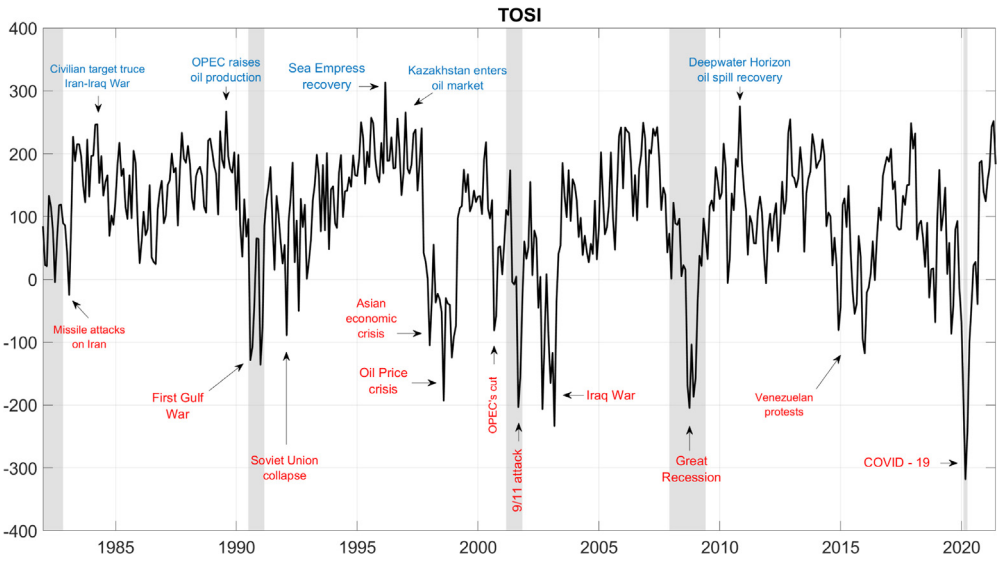


Fig. 3. Common component extracted from a dataset comprising VADER and BERT text indicators. Blue and red notes describe historical events that positively or negatively impacted oil prices. Shaded areas represent recessions as identified by the NBER. The sample period covers January 1982 to June 2021, with the index normalised to a mean of 100. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 3 plots TOSI over the period January 1982 - June 2021. The index clearly tracks the major geopolitical events that impact oil prices. In particular, the Iraq War, the COVID-19 pandemic, and the global financial crisis are correctly identified as the main factors that negatively affected the global economy and consequently the price of oil. Additionally, in the range around $[-100; -200]$ on the index scale, one can also identify earlier negative shocks, such as the Gulf war, the oil price crisis of 1998, where Brent price fell below £18 per barrel, the Twin Towers attack and the Venezuelan protests of 2016 demanding President Maduro resign. On the other hand, TOSI rises during periods of market stabilisation following increased oil production, reflecting positive supply-side developments that supported both WTI and Brent prices.

To evaluate TOSI's predictive power, I estimate the SVBVAR from Eq. (5) with TOSI included as an endogenous variable. Importantly, the common factor is updated recursively in real time, so that at each forecast date, only information available up to that point is used. For reliability, I compare TOSI's performance against Abiad and Qureshi (2023)'s oil price uncertainty indicator (OPU) and Baker and Wurgler (2006)'s investor sentiment measure (BW). The former also draws on global newspaper text. These comparisons underscore a key contribution of this paper: not only do text-based sentiment indicators outperform text-based uncertainty measures in forecasting real oil prices, but TOSI is specifically tailored to oil market dynamics rather than general market sentiment.

Panel A in Table 5 presents the recursive MSPE ratios for each model compared to a random walk forecast. The results indicate that TOSI outperforms the other indices, with statistically significant improvements at the 1% level in the short term. Notably, OPU-based forecasts show that the uncertainty-driven tf-idf index from Section 3 performs slightly better in short-term forecasts and comparably in medium-term forecasts, despite being derived from a single data source. These findings further support the notion that sentiment-based text indicators yield more accurate oil price forecasts than uncertainty-based measures. Furthermore, the TOSI-based SVBVAR model also outperforms the equivalent model without text data across all forecast horizons and for every oil price measure. In particular, the improvements relative to the no-text model are statistically significant in the short, medium and long term.

Panel B of Table 5 applies the Model Confidence Set (MCS) procedure by Hansen et al. (2011) to assess the predictive power of all models jointly. Consistent with Panel A, the SVBVAR model with TOSI emerges as the top performer, demonstrating statistically superior short-term predictability.

Analysing the cumulative sum of prediction errors for a TOSI-based SVBVAR model, it is evident that this text index consistently yields more accurate forecasts than a random walk (top row of Fig. 4) and the benchmark model in Eq. (5) (bottom row of Fig. 4), particularly in the short and long term. Furthermore, during periods of high volatility, such as the global financial crisis and the COVID-19 pandemic, when accurate predictions are critical, TOSI significantly enhances point forecast accuracy.

Further robustness experiments, detailed in Appendix G, explore two additional factor-extraction techniques (dynamic factor VAR and the three-pass regression filter) as well as alternative SVBVAR specifications in which text indices enter as instrumental variables. These exercises also demonstrate that TOSI is not simply a proxy for general sentiment or uncertainty. When examined

Table 5
Recursive MSPE (Panel A) and MCS (Panel B) of alternative monthly indicators of real oil prices in a SVBVAR(12). Comparison between different predictors.

Panel A: MSPE	h = 1		h = 3		h = 6		h = 12		h = 24	
Predictor	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
TOSI	**0.838*****	**0.829*****	**0.858**	**0.884**	**0.899**	**0.897**	**0.954**	**0.937**	**0.916**	**0.900**
OPU	0.887**	0.881**	0.929	0.895	0.958	0.928	0.970	0.959	0.919	0.904
BW	0.884**	0.898**	0.918	0.928	0.955	0.945	1.009	0.993	0.977	0.988

Panel B: MCS	h = 1		h = 3		h = 6		h = 12		h = 24	
Predictor	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
TOSI	**1.000**	**1.000***	**1.000**	0.993	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
OPU	0.255	0.158	0.460	**0.996**	0.578	0.917	0.996	0.990	1.000	1.000
BW	0.675	0.167	0.657	0.703	0.818	0.910	0.983	0.940	0.983	0.568

Note: Panel A displays MSPE ratios relative to a random walk model. Bold values indicate the lowest MSPE results relative to a specific time horizon h and oil price measure (WTI and Brent). *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively, when tested against the random walk model, while blue, red, and yellow cells denote significance levels of 10%, 5%, and 1%, respectively, when tested against the No-text-based SVBVAR model. Significance tests are performed using the Diebold–Mariano test. In Panel B, bold values highlight models included in the Superior Set of Models, as identified by the MCS procedure at the 1% confidence level. *, **, *** respectively denote 10%, 5% and 1% level of significance, for a test of the null hypothesis that there is no difference in predictive accuracy among the models.

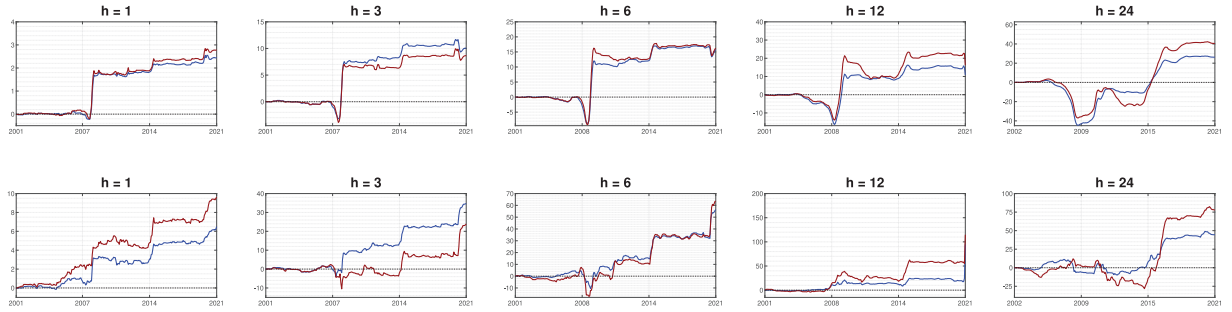


Fig. 4. Top row plots the difference between the cumulative sum of forecasting errors of Eq. (5) with TOSI endogenised and a random walk. Bottom row plots the difference between the cumulative sum of forecasting errors of Eq. (5) with and without TOSI. In both cases, VARs are estimated in a Bayesian fashion by assuming stochastic volatility in the error term. The blue and red lines represent WTI and Brent crude oil prices, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

during periods of pronounced oil-market stress, TOSI exhibits only modest correlation with broader sentiment and uncertainty measures yet responds far more sharply and precisely to oil-specific shocks. This distinction is evident both in graphical analyses and in out-of-sample forecasting exercises over recent disruptive events, where the TOSI-enhanced model consistently achieves lower MSPEs than its alternatives. Collectively, these findings confirm that TOSI effectively captures the nuances of oil-market sentiment, reinforcing its value as a targeted and real-time forecasting indicator.

5.1. TOSI evaluation: point and density accuracy

Having shown that TOSI delivers superior out-of- sample forecasts in a full SVBVAR framework, I now isolate its incremental value by comparing nested models of increasing dimensionality, both with and without the text indicator.

Specifically, I begin with a univariate autoregression model where the price of oil is the sole variable. Then I compare the forecasting outcomes of this model to those of a 2-variable SVBVAR, where TOSI is considered as the second endogenous variable. Subsequently, I run a 2-variable SVBVAR with oil prices and the global real economy, and compare the forecasting outcomes to those of a 3-variable SVBVAR with TOSI endogenised. Next, I run a 3-variable SVBVAR with oil prices, the global real economy and oil production, comparing it to a 4-variable SVBVAR with TOSI endogenised. Finally, I incorporate oil inventories and replicate the previous exercise. Variable selection follows a systematic approach, prioritising variables released earliest and concluding with those made available with greater delay.

Fig. 5 shows that, across all horizons ($h = 1, 3, 6$ months), endogenising TOSI consistently reduces point-forecast error, often at statistically significant levels. The purpose of this experiment is twofold. Firstly, it demonstrates the robustness of the forecasting gains obtained

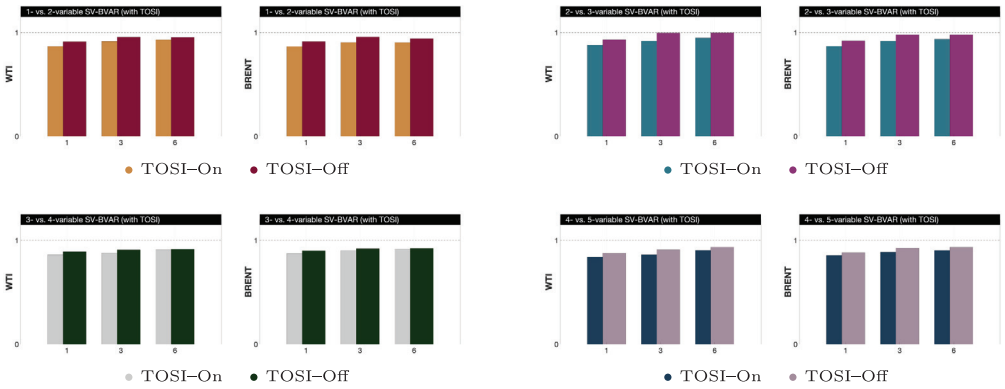


Fig. 5. Comparison between MSPEs among four alternative SVBVAR models with and without TOSI. The oil measures are reported on the *y*-axis, while the month-ahead forecast is displayed on the *x*-axis. Models beat the random walk when bars lie below 1.

using TOSI in different model specifications. Secondly, it indicates that despite the delayed availability of oil market drivers, text data consistently yields improved forecasts. Importantly, since text information does not require any revision, TOSI ensures reliable performance over time.

In the second experiment, I assess whether TOSI also enhances density forecasts across alternative model specifications. To this hand, I use the averages of log predictive likelihoods (ALPL) and the average continuous ranked probability score (ACRPS) to evaluate the quality of the predictive densities. The ALPL is recognised as a comprehensive metric for evaluating the overall fit of the predictive distribution to observed data (Geweke & Amisano, 2010). Higher ALPL scores indicate better probability density forecasts. On the other hand, the ACRPS offers a nuanced evaluation by giving greater emphasis to forecast values that are close to, but not necessarily equal to, the actual outcome. This makes it less susceptible to the influence of outliers, providing a more balanced measure of forecast performance (Gneiting & Raftery, 2007; Gneiting & Ranjan, 2011). For ACRPS, the closer the score is to zero, the more accurate the forecast is. The performance of TOSI-based SVBVAR models is compared to both random walk and SVBVAR models without the text indicator for the forecast horizon $h = 1, 3, 6, 12$ and 24.

As shown in Table 6, the accuracy of short- and medium-term density forecasts significantly improves when TOSI is incorporated into a 5-variable SVBVAR model. Specifically, Panel A highlights that the ALPL exhibits statistically significant gains at the 1% level compared to the no-change forecast across all time horizons and oil price measures. In some instances, the results also show significance at the 5% and 10% levels when compared to equivalent models that do not include TOSI. Turning to Panel B, which reports the ACRPS, we observe that, except for Brent crude oil prices, there are statistically significant improvements at the 1% level compared to a random walk model, extending up to a 12-month forecast for WTI. The ACRPS reveals more cases where incorporating TOSI leads to better performance compared to models without TOSI. In particular, for WTI forecasts, there is a 1% statistically significant improvement from the first month up to a 12-month horizon, and a 10% improvement for a 24-month horizon. For Brent crude oil prices, we observe a 10% significance level improvement for the 1- and 6-month horizons. Consistent with the point forecast analysis, the inclusion of text data results in lower but still statistically significant gains for long-term forecasts. For expository purposes, evidence of SVBVAR models without text data is not reported in Table 6, although this was investigated. For 1- and 2-year ahead forecasts, incorporating TOSI into a 2-variable SVBVAR model yields the most accurate results, marginally outperforming the equivalent model that excludes TOSI. Since TOSI data are available in real time, these findings are particularly valuable. They suggest that forecasters can rely on this new driver to improve long-term oil price density forecasts when other oil market fundamentals are not immediately available.

Overall, these findings align with the outcomes of previous experiments and strongly support the integration of TOSI into practical econometric models for forecasting real oil prices.

5.2. On the directional accuracy of oil price forecasts

In the previous sections, I examined the impact of TOSI on forecasting crude oil spot prices over different horizons. The accuracy of out-of-sample forecasts was evaluated by comparing the forecasted distribution's central tendency with the actual observed values. Forecasts that deviated significantly from the realised values resulted in higher prediction errors. However, central banks often prioritise accurate directional predictions over mere proximity to actual values (Alquist & Kilian, 2010; Baumeister & Kilian, 2016). This focus helps them anticipate inflationary pressures and adjust interest rates accordingly. For this reason, it is important to determine whether text data enhances the directional accuracy of oil price forecasts.

In this section, I analyse the directional accuracy of short-term real oil price forecasts using relative operating characteristic (ROC) curves. The ROC curve methodology determines the directional accuracy by comparing true

Table 6
Recursive ALPL (Panel A) and ACRPS (Panel B) of alternative monthly indicators of real oil prices across various SVBVAR(12) models.

Panel A: ALPL		h = 1		h = 3		h = 6		h = 12		h = 24	
Variables	Model	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
p^{oil}, *TOSI*	2-SVBVAR	1.021***	**0.143***	0.206***	-0.102***	-0.168***	-0.377***	**-0.482***	-0.643***	**-0.777***	**-0.945***
GCOP, p^{oil}, *TOSI*	3-SVBVAR	1.016***	0.122***	0.202***	-0.124***	-0.169***	-0.397***	-0.488***	-0.653***	-0.784***	-0.956***
GCOP, *WIP*, p^{oil}, *TOSI*	4-SVBVAR	1.018***	0.084***	0.226***	-0.146***	-0.159***	-0.418***	-0.508***	-0.692***	-0.815***	-0.987***
GCOP, *WIP*, *Oinv*, p^{oil}, *TOSI*	5-SVBVAR	**1.030***	0.099***	**0.245***	**-0.075***	**-0.144***	**-0.341***	-0.492***	**-0.634***	-0.812***	-0.973***

Panel B: ACRPS		h = 1		h = 3		h = 6		h = 12		h = 24	
Variables	Model	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT	WTI	BRENT
p^{oil}, *TOSI*	2-SVBVAR	0.062***	0.114	0.129***	0.171	0.182***	0.226	**0.247***	**0.296**	**0.307***	0.374
GCOP, p^{oil}, *TOSI*	3-SVBVAR	0.062***	0.115	0.130***	0.174	0.183***	0.230	0.248***	0.301	0.312	0.384
GCOP, *WIP*, p^{oil}, *TOSI*	4-SVBVAR	0.061***	0.116	0.128***	0.176	0.182***	0.233	0.256	0.311	0.328	0.398
GCOP, *WIP*, *Oinv*, p^{oil}, *TOSI*	5-SVBVAR	**0.060***	**0.112**	**0.125***	**0.167**	**0.176***	**0.219**	0.248***	**0.296**	0.308*	**0.371**

Note: In column 1: p^{oil} = oil prices, *GCOP* = global crude oil production, *WIP* = world industrial production, *Oinv* = oil inventories, *TOSI* = text oil sentiment indicator. For negative results, a less negative value indicates that the model offers a more accurate representation of the true density forecast. Bold values indicate the best model relative to a specific time horizon h and oil price measure (WTI and Brent). *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively, when tested against the random walk model, while blue, red, and yellow cells denote significance levels of 10%, 5%, and 1%, respectively, when tested against the equivalent model without TOSI. Significance tests are performed using the Diebold–Mariano test.

positive (*TP*) rates with false positive (*FP*) rates for alternative values of a decision criterion c (Berge & Jordà, 2011; Lahiri & Wang, 2013). For this experiment, I define $p_t^{oil} \in \{0, 1\}$ as the actual value of Brent crude oil prices, with 0 and 1 respectively denoting increasing and decreasing prices at time t. Real-time probability forecast of oil prices is denoted as $\hat{p}_t^{oil}$. An increase in oil prices is predicted when $\hat{p}_t^{oil} \geq c$, and a decrease is predicted when $\hat{p}_t^{oil} < c$. Based on these definitions, the following conditional probabilities are determined:

$$TP(c) = P\left[\hat{p}_t^{oil} \geq c | p_t^{oil} = 0\right], \tag{7}$$

$$FP(c) = P\left[\hat{p}_t^{oil} \geq c | p_t^{oil} = 1\right]. \tag{8}$$

The ROC curve summarises the set of possible combinations of TP(c) and FP(c) for $c \in (-\infty, +\infty)$ within a $[0, 1] \times [0, 1]$ space. The curve lies above the 45-degree bisecting line if the forecasts outperform a random noise signal. Three experiments are conducted in this analysis. In the first, I compare the ROC curves generated by SVBVAR models with and without the inclusion of TOSI. For robustness, I use either Baumeister and Hamilton (2019)'s world industrial production index (WIP) and Kilian (2009)'s index (REA) as a proxy for global real economic activity. The results, depicted in the top panel of Fig. 6, indicate that TOSI-based SVBVAR models have superior directional accuracy in forecasting short-term Brent crude oil prices. Specifically, models that exclude text data generate ROC curves that are close to or below the bisecting line, suggesting that the information provided by these models is comparable to or worse than that of a random noise signal.

Next, I estimate the area under the ROC curve (AUROC) using a rolling-estimation window of 48 months. An AUROC of 1 indicates perfect prediction accuracy, whereas an AUROC of 0.5 suggests predictions no better than random chance. If predictions consistently forecast the incorrect direction of price changes, then AUROC will be less than 0.5. The results, reported in the middle and bottom panels of Fig. 6, reveal that the inclusion of TOSI positively impacts the directional accuracy of short-term oil price forecasts over time, as the AUROC values of the TOSI-based models are consistently higher than those of the models without text data. Additionally, TOSI-based AUROC scores often exceed 0.5, indicating that forecasts incorporating text data outperform random noise signals. Similar conclusions cannot be claimed for models without text data.

In the last exercise, I report the AUROC estimates resulting from Brent crude oil price short-term forecasts across the full sample and during periods of economic recessions in the Eurozone, as determined by the CEPR-EABCN Euro Area Business Cycle Dating Committee. As shown in Table 7, the results for the various subsamples reinforce the full-sample results. Specifically, for all periods of economic recession, a TOSI-based SVBVAR model provides better predictive performance in forecasting the sign of changes in Brent crude oil prices. AUROC scores are also tested under the null hypothesis that oil price forecasts are indistinguishable from a pure noise signal. The evidence shows that TOSI-based AUROC estimates are statistically significant at the 10% level during the COVID-19 period and at the 1% level during the global financial crisis and the full sample period. This suggests that the TOSI index provides valuable insights for predicting both the mean of the distribution and the direction of subsequent changes in crude oil prices.

In summary, 5-variable SVBVAR models with endogenous text variables outperform all other models investigated. Moreover, combining multiple text sources helps minimise bias and significantly enhances oil forecasts. The Financial Times and Thomson Reuters, on average, incorporate more useful information for predicting WTI and

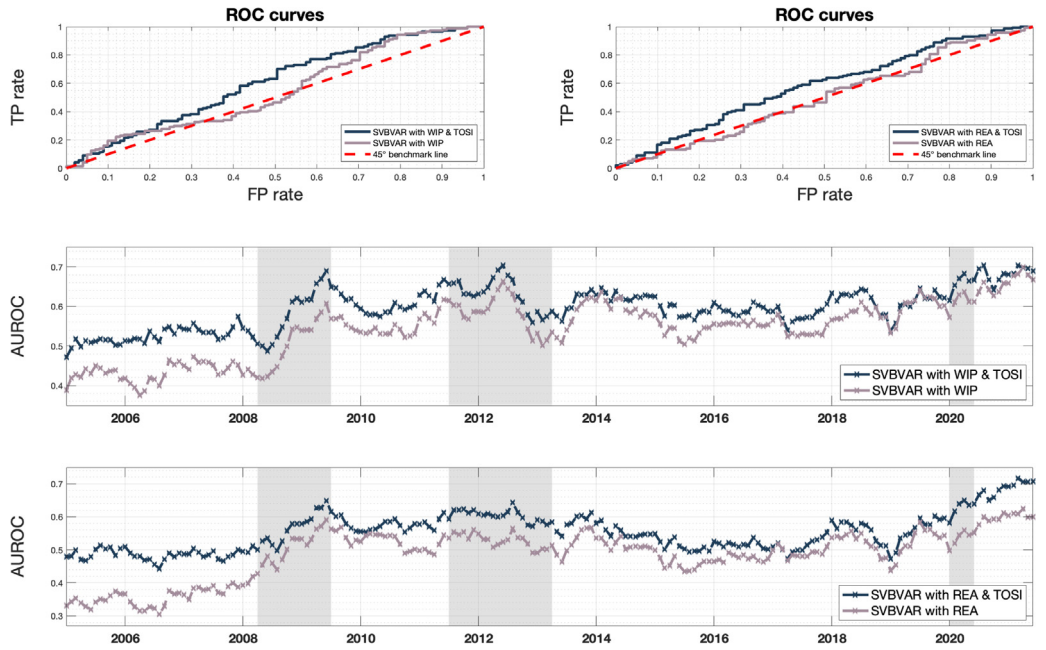


Fig. 6. Top panel compares the ROC curves for Brent crude oil 1-month ahead forecasts, with and without incorporating TOSI, using two different indicators of global real economic activity: the world industrial production Index (WIP) and the Kilian index (REA). Red benchmark lines denote a no-information scenario, with forecasts providing no directional accuracy regarding future oil prices. The middle and lower panels display rolling-window estimates of the Area Under the ROC Curve (AUROC) statistic for Brent crude oil 1-month ahead forecasts. Higher values indicate better predictive performance. The shaded areas indicate recessions as identified by the CEPR-EABCN Business Cycle Dating Committee. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 7
AUROC estimates.

Economic period	Time frame	4-SVBVAR	TOSI-based 5-SVBVAR
Global financial crisis	2008M4 – 2009M7	0.7857**	**0.8036*****
Eurozone sovereign debt crisis	2011M7 – 2013M4	0.4636	**0.5909**
COVID-19 recession	2020M1 – 2020M6	0.6250	**0.8750****
Full sample	1982M1 – 2021M6	0.5436	**0.6044*****

Note: Recession dates are determined by the CEPR-EABCN Euro Area Business Cycle Dating Committee. Bold values indicate superior predictive performance. *, **, and *** respectively denote 10%, 5%, and 1% level of significance, for a test of the null hypothesis that oil price forecasts are indistinguishable from a pure noise signal.

Brent oil prices. The BERT metric provides more useful information for 1-, 3-, and 6-month ahead forecasts, while VADER performs best for the 24-month horizon. The first principal component extracted from the BERT and VADER indices in real time improves both distribution forecasts and directional accuracy of real oil prices. These results hold for both point and density forecasts.

Many additional robustness checks have been conducted, with further details provided in the appendix. Specifically, I have replaced the world industrial production indicator with both Baumeister et al. (2022)'s GECON indicator and Kilian (2009)'s REA index, replicating previous experiments. Next, I have also substituted global oil production with a measure of world oil consumption. Furthermore, I have explored the combined use of GECON (or REA) and oil consumption as alternatives to the standard indicators. Experiments on refiner acquisition cost (RAC) for U.S. oil import prices have also been performed, and the results align with those presented above, further demonstrating the clear benefits of integrating TOSI into monthly real-oil-price forecasting.

6. Conclusions

This work introduces a set of text-based time series variables assessing (i) oil market sentiment and (ii) oil price uncertainty. Implementing these measures in alternative vector autoregression models reveals both strengths and weaknesses of textual data. The weaknesses are mainly associated with uncertainty measures, which often fail to respond to key geopolitical events that affect oil prices, consequently leading to inaccurate forecasts of various oil price measures. In contrast, sentiment indicators can effectively track historical events that impact oil prices, thereby enabling econometric models that rely on such indicators to improve the forecasting accuracy of real oil prices significantly.

The evidence also shows that endogenising sentiment-based text regressors in an SVBVAR model results in forecast performance of the econometric model not only outperforming the random walk in the short, medium, and long term, but the MSPEs associated with such text data are also significantly lower than those of the benchmark model commonly used in the literature. The first principal component extracted from the best-performing text indicators yields a new text-based index that significantly enhances real oil price point and density forecasts, especially during periods of economic instability.

In conclusion, this work provides cutting-edge guidance for energy companies, international organisations, and, in particular, central banks that require accurate forecasts of real oil prices to inform policy decisions.

CRediT authorship contribution statement

Luigi Gifuni: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.09.001.

Data and code availability

The data and code to reproduce the results in this paper are available at https://github.com/gifuniluigi/ijof-wom_esui.

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